Noise Analysis

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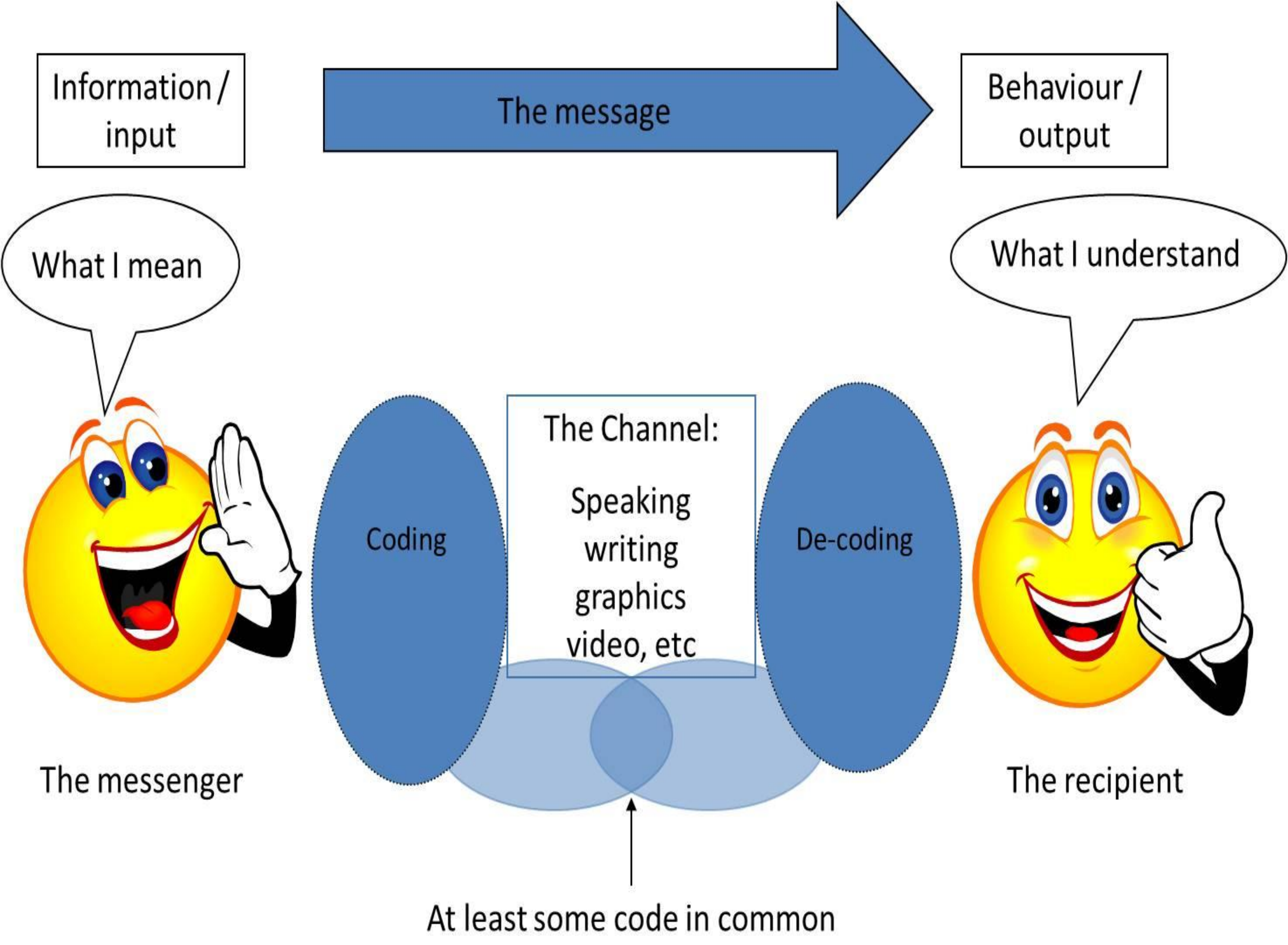
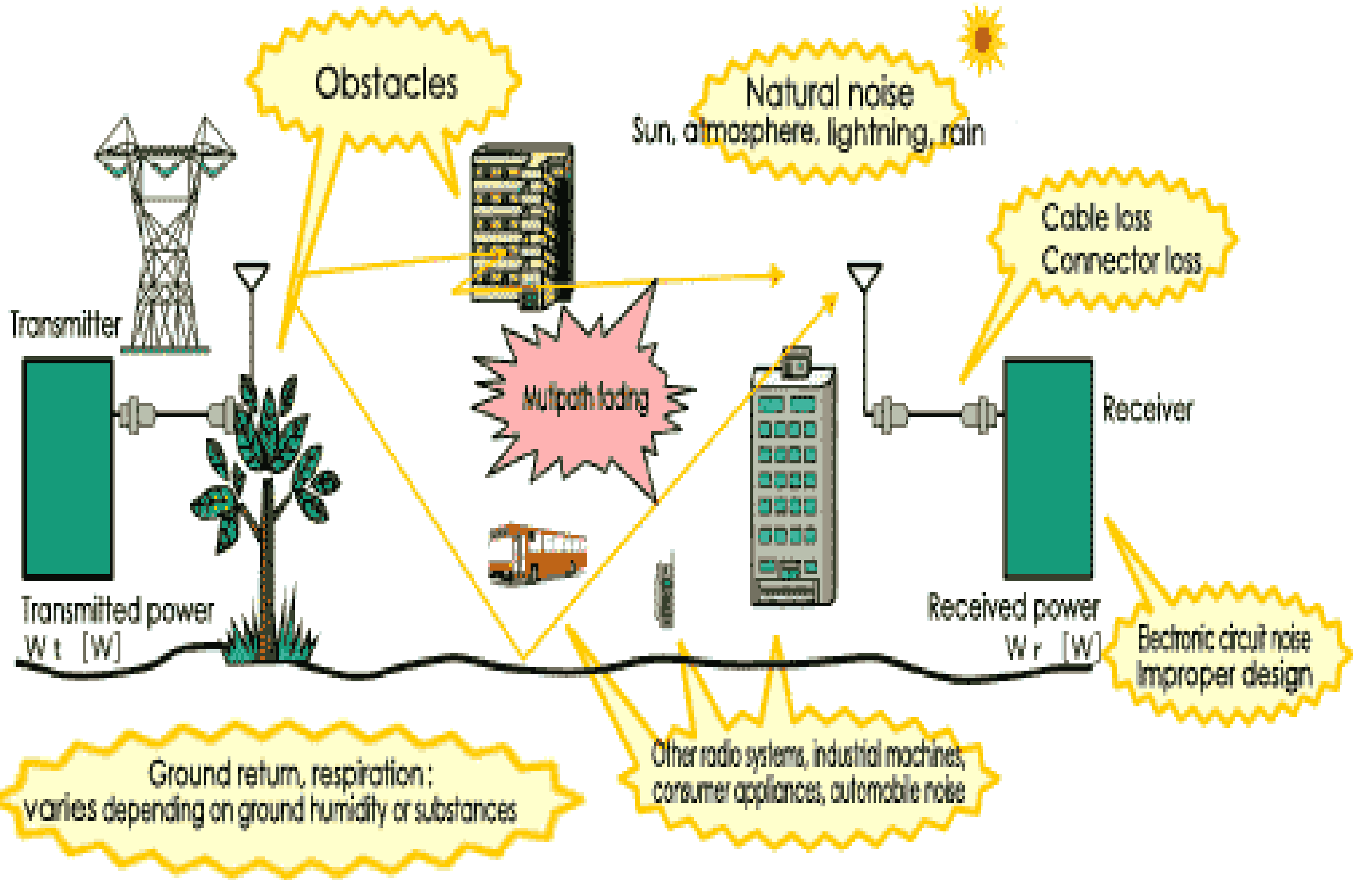
Noise

Noise is a physical barrier to effective communication. Noise may have its origin from an external source or may exist even in the communication loop.


Information / input
The message
Behaviour / output
What I mean
What I understand
Coding
The Channel:
Speaking
writing
graphics
video, etc
De-coding
The messenger
The recipient
At least some code in common

Noise and loss that may cause errors



Definition of Noise

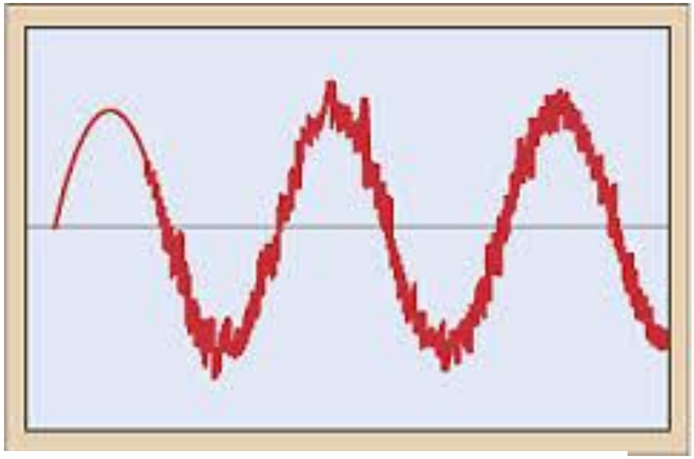
- Noise refers to any unwanted signal that interferes with the transmission of information

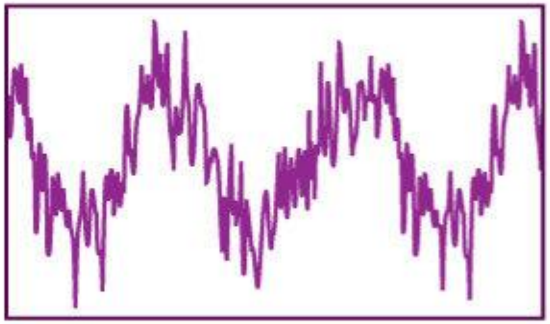
- It degrades the quality of the transmitted signal, making it difficult for the receiver to interpret the information correctly

- Noise sources→ electronic components, environmental factors, and external interference

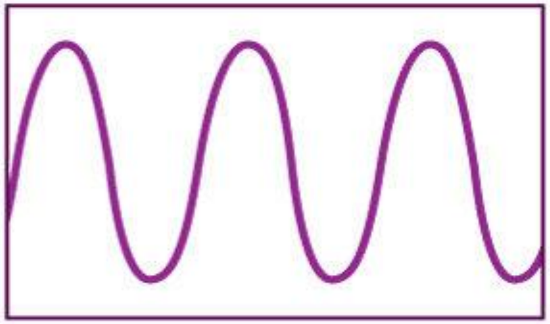
Electrical Noise

"It is undesirable electrical energy which falls in the pass band of the signal, which gives rise to audible noise in a system."

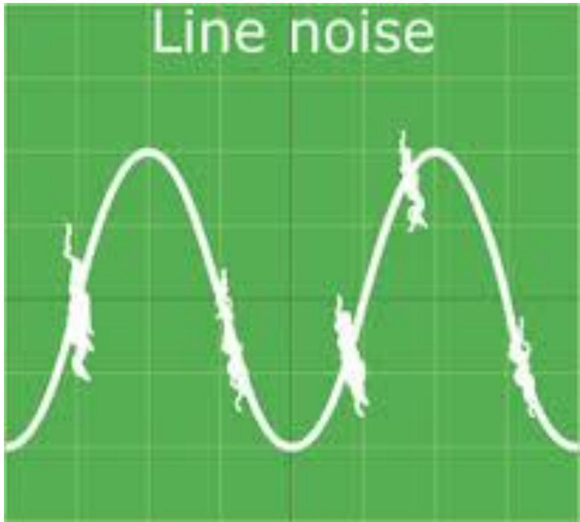


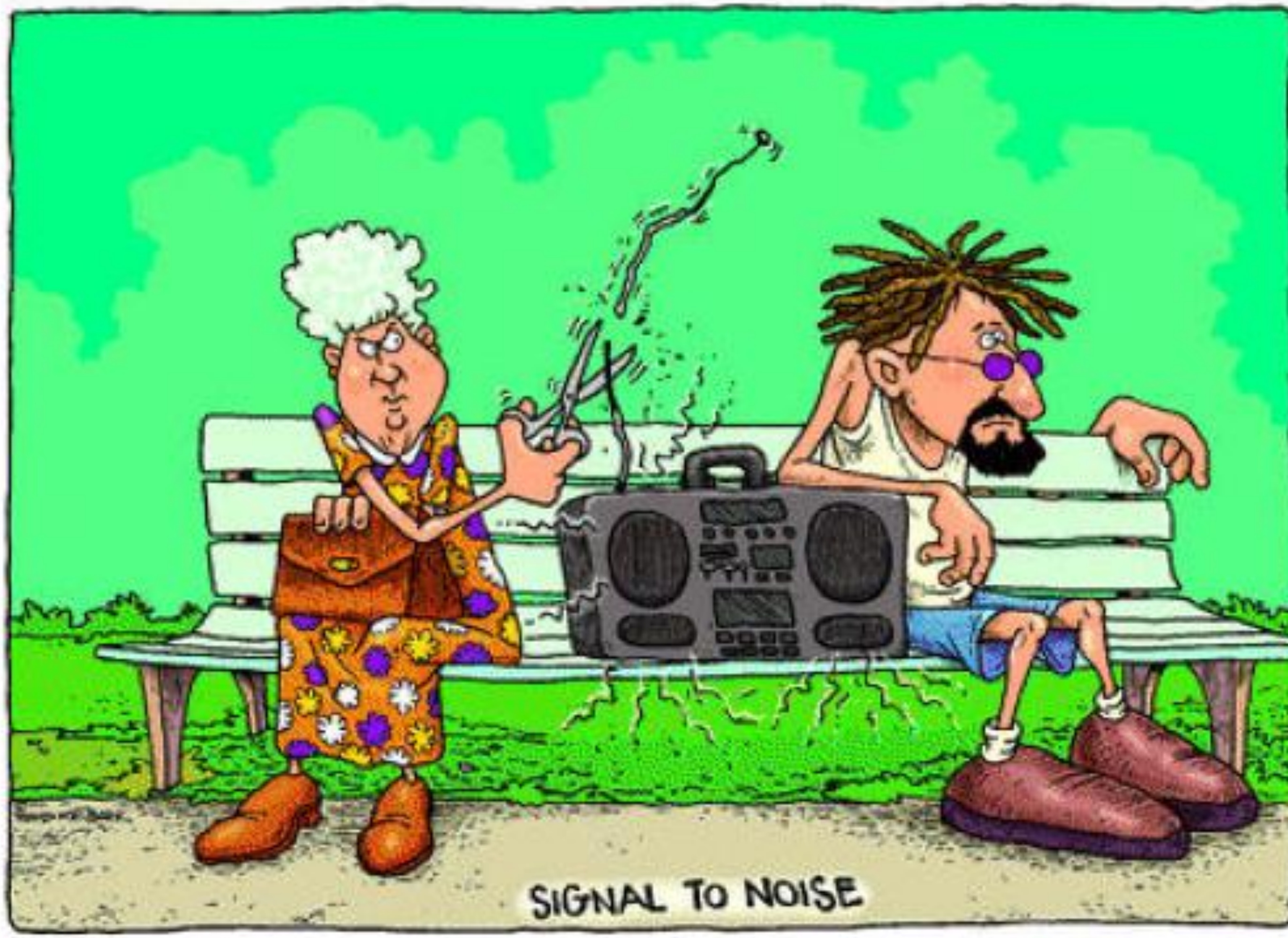


Regular AC flow



Pure power AC flow





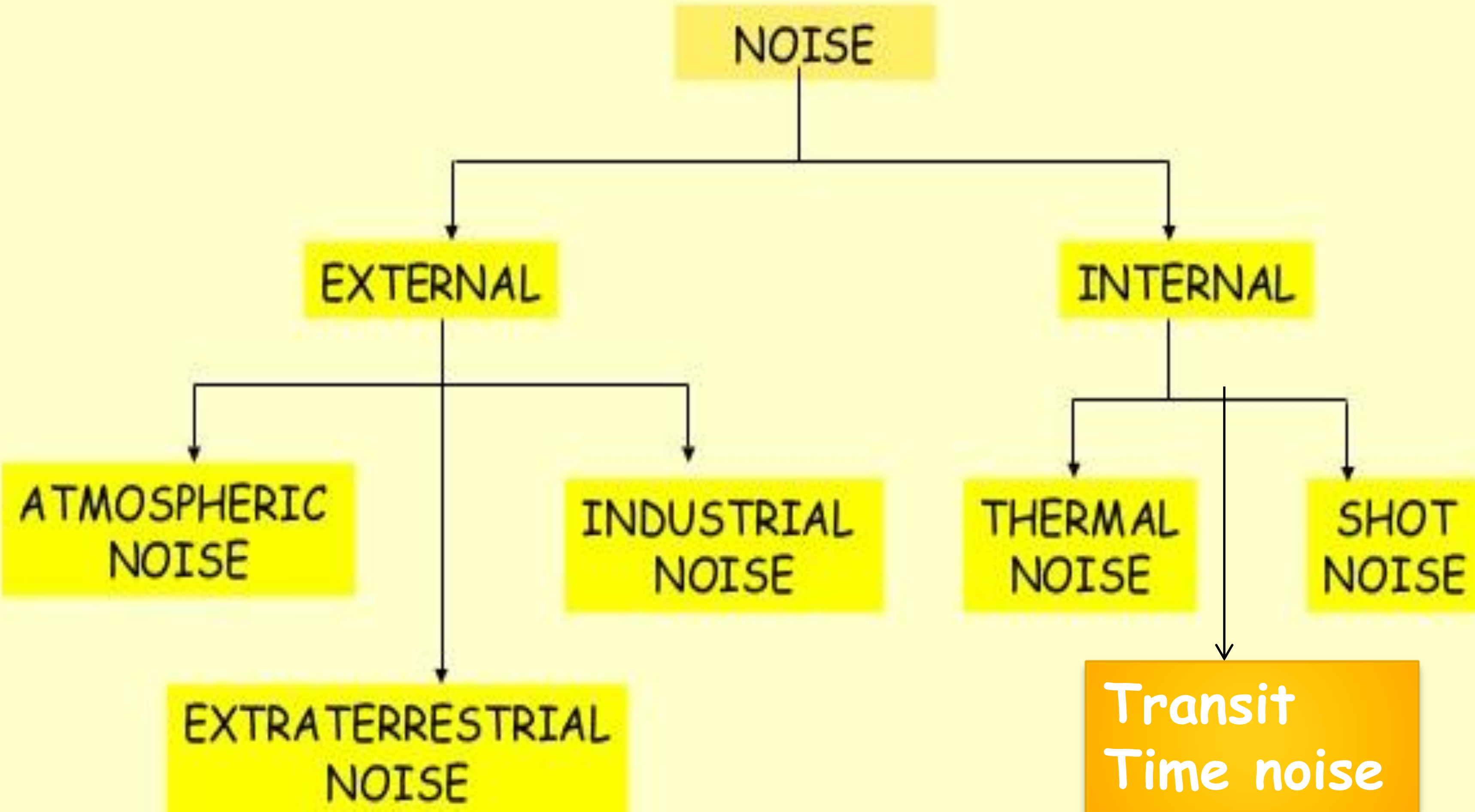
"Whether noise is a nuisance or a signal may depend on whom you ask"

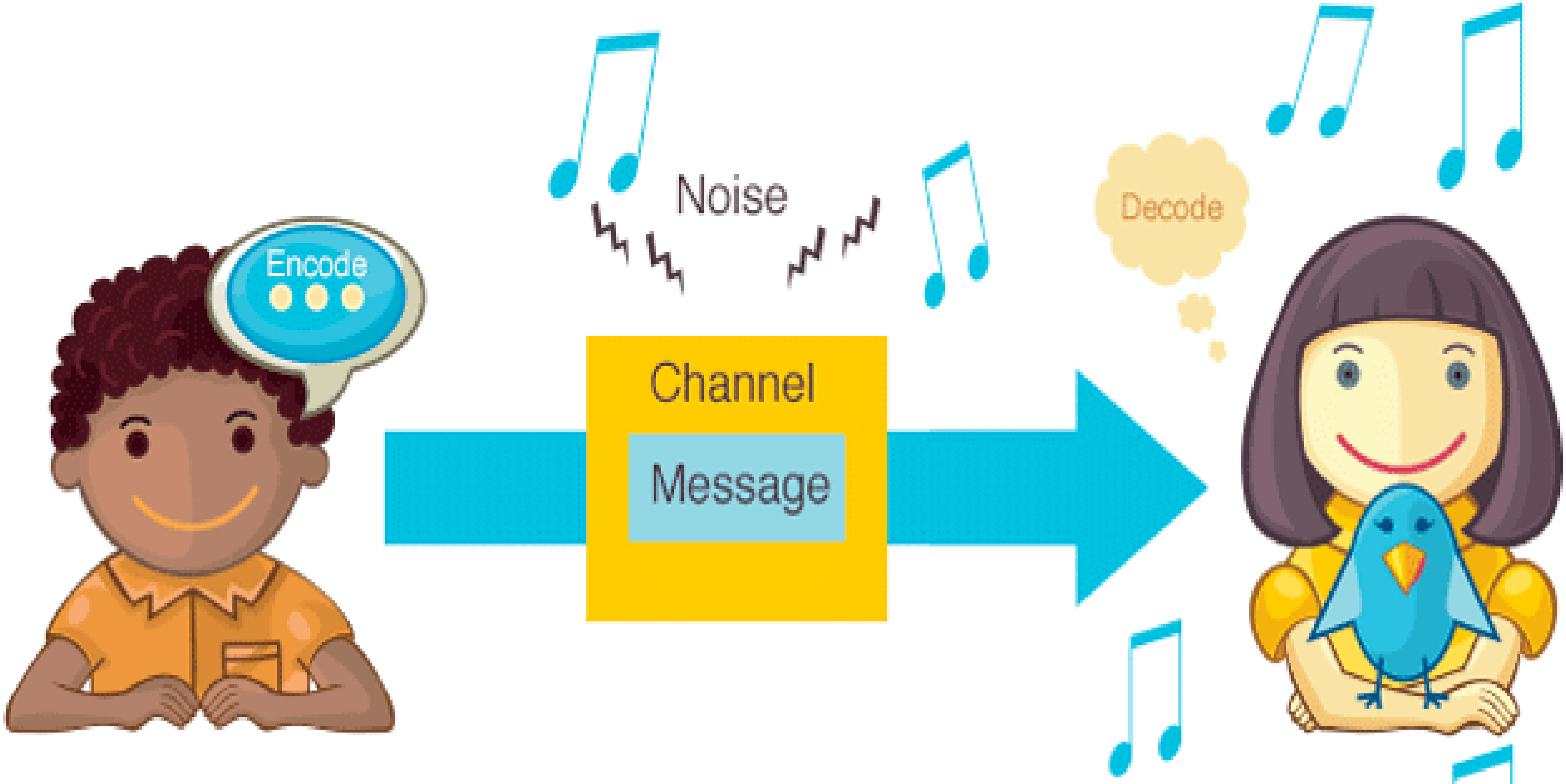
Classification of Noise

- Correlated-Exists only when signal is present

- Uncorrelated-Exists all the time

Classification of Uncorrelated Noise





Bird chirping = external noise

INTERNAL NOISE

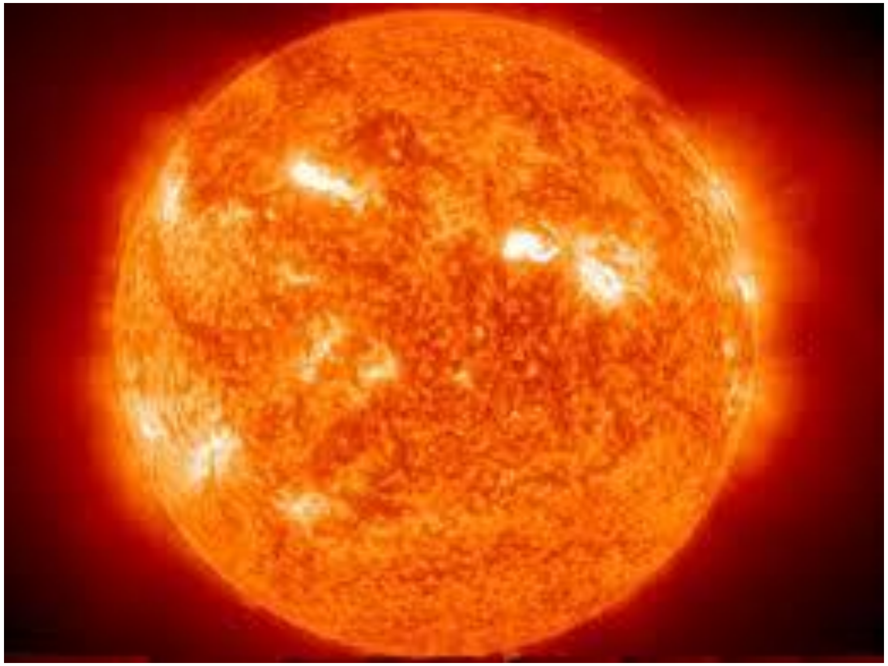
The noise generated within the electronic devices or circuits is called as internal noise.

Atmospheric Noise

- Naturally occurring electrical disturbances from earths atmosphere.
- Sputtering, crackling and comes from speaker when there is no signal.
- Source of static electricity is lightning (spread energy throughout wide range of frequency)

Extraterrestrial Noise

- Electrical signals that originates outside earths atmosphere.
- Called as deep space noise
- Originates from milkyway, other galaxies and sun



Extraterrestrial Noise –Solar Noise

- Generated directly from sun's heat.

- A quiet condition when relative constant radiation.

- High intensity caused by sun spot activity and solar flare-ups.

Extraterrestrial Noise –Cosmic Noise

- Sources of this noise are distributed throughout the galaxies.

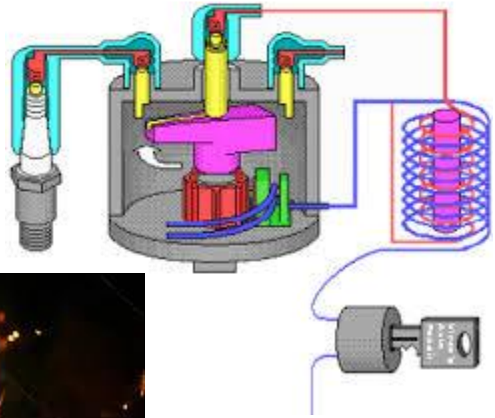
- Often called as black body noise.

Industrial (Man made) Noise

- Sources are spark producing mechanism such as motors, automobile ignition systems, AC power generating and switching equipment. Fluorescent light.

- Contains wide range of frequencies.

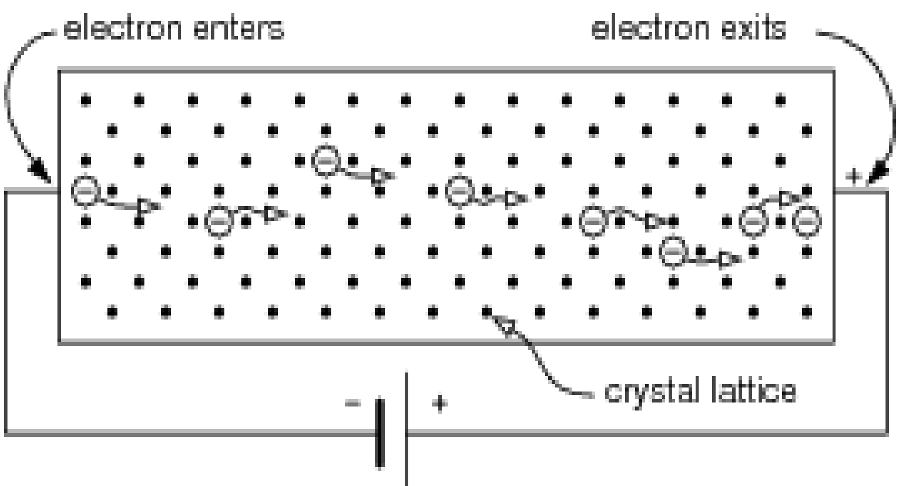
- More intense and more dense in metropolitan and industrial areas



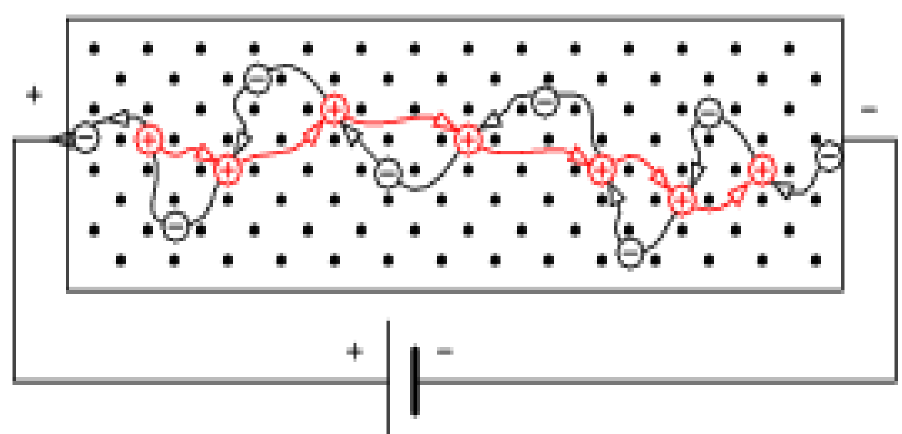
Internal Noise- Shot noise

- Caused by random arrival of carriers (holes and electrons) at the output element.

- Carriers are not moving in a continuous steady flow, and hence distance vary.



(a) N-type



(b) P-type

Internal Noise- Transit-time noise

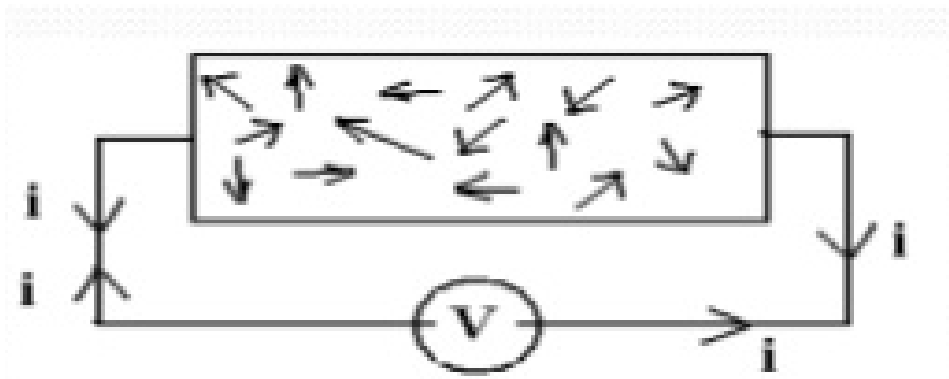
- Modification to a stream of carriers as they pass from input to output.

- This noise is determined by carrier mobility, bias voltage, and transistor construction.

- Carriers suffer from emitter time delays, base transit time delay and collector recombination time.

Internal Noise- Thermal Noise

- Rapid and random movement of electrons within conductor due to thermal agitation.



- Because electron movement is random it is called as random noise.

- White noise because it is present at all frequency

Internal Noise- Thermal Noise (Johnson-Nyquist Noise)

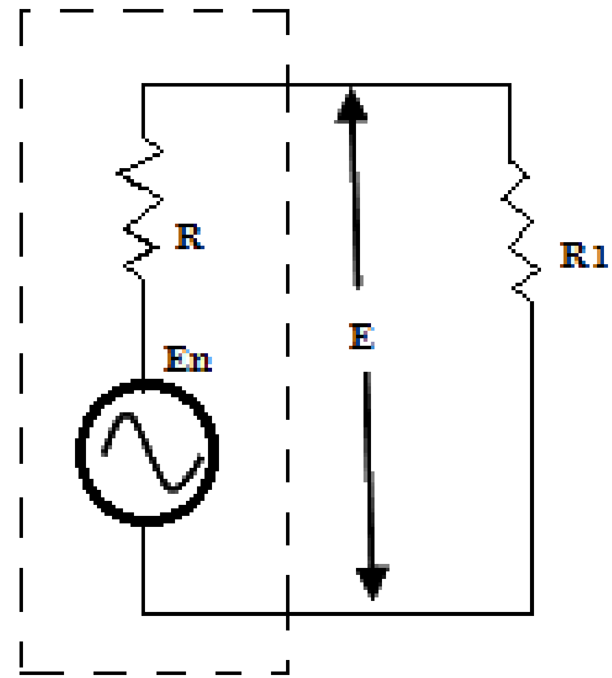
- Johnson proved that thermal noise power is proportional to the product of bandwidth and temperature.

$$P_n = \mathrm{KTB}$$

- P_n= Noise power
- B= Bandwidth
- K= Boltzman's constant ($1.38 \times 10^{-23}\ Joules/Kelvin$)
- T= Absolute temperature (kelvin)

Thermal Noise- Noise Voltage

$$P_n = \frac{E^2}{R_1} = \frac{E^2}{R}$$



$$P_n = \frac{\left(\frac{E_n}{2}\right)^2}{R} = \frac{{E_n}^2}{4R}$$

$${E_n}^2 = 4RP_n = 4RKTB$$

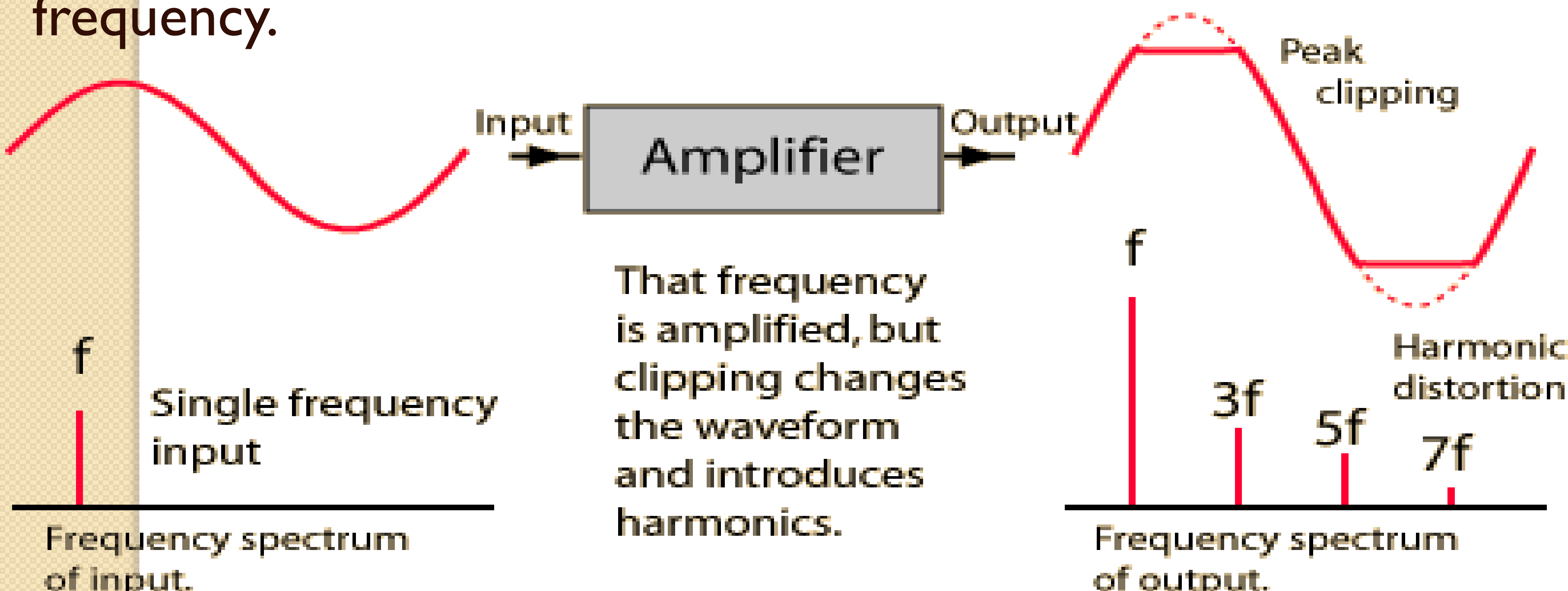
$$RMS\ Noise\ Voltage \rightarrow E_n = \sqrt{4RKTB}$$ **(volts)**

Correlated Noise- Harmonic Distortion

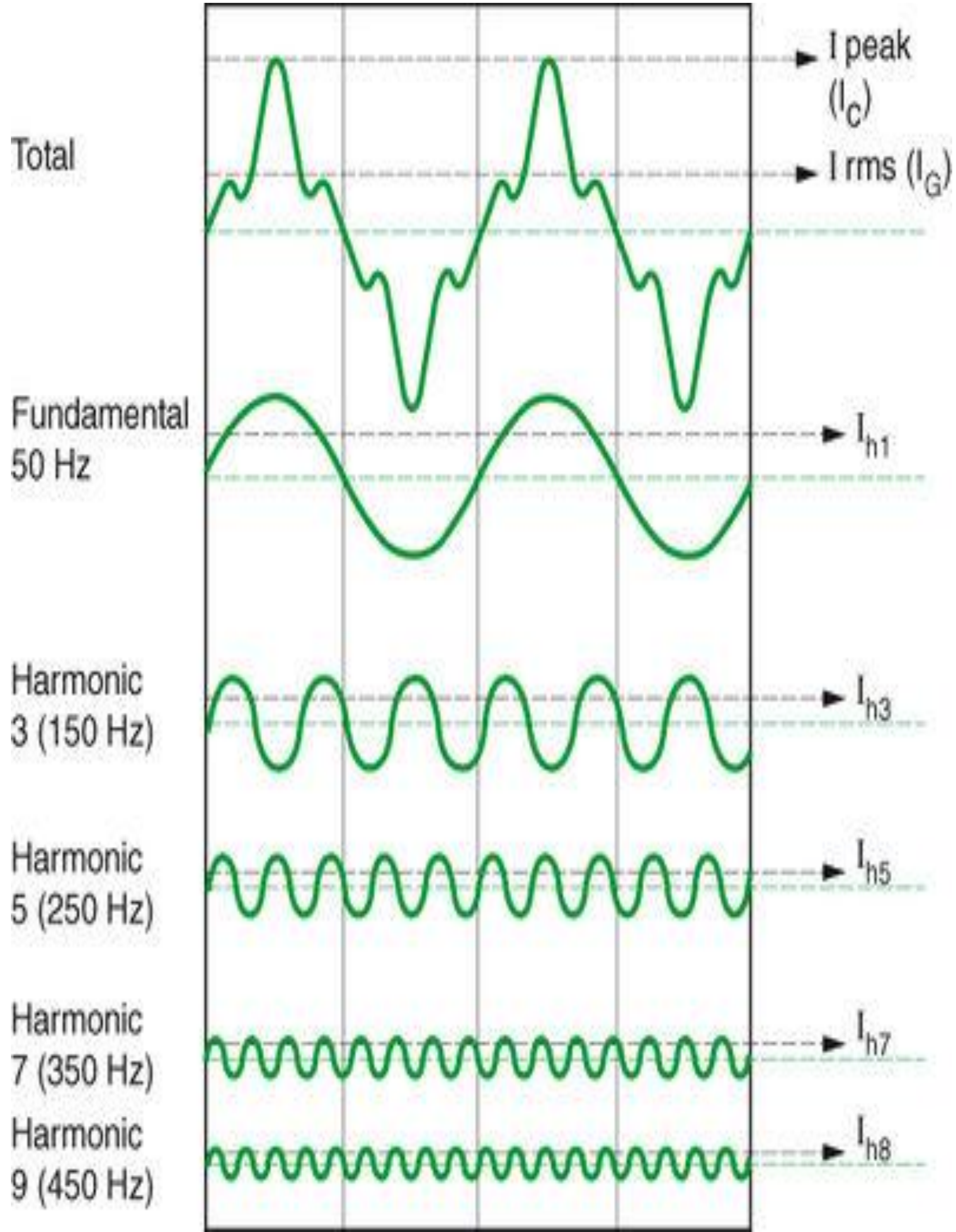
- Occurs when harmonics of a signal are produced due to nonlinear amplification.

- Harmonics are integer multiple of the original signal.

- The original signal is first harmonics and called as fundamental frequency.



Correlated Noise- Harmonic Distortion



Correlated Noise- Harmonic Distortion

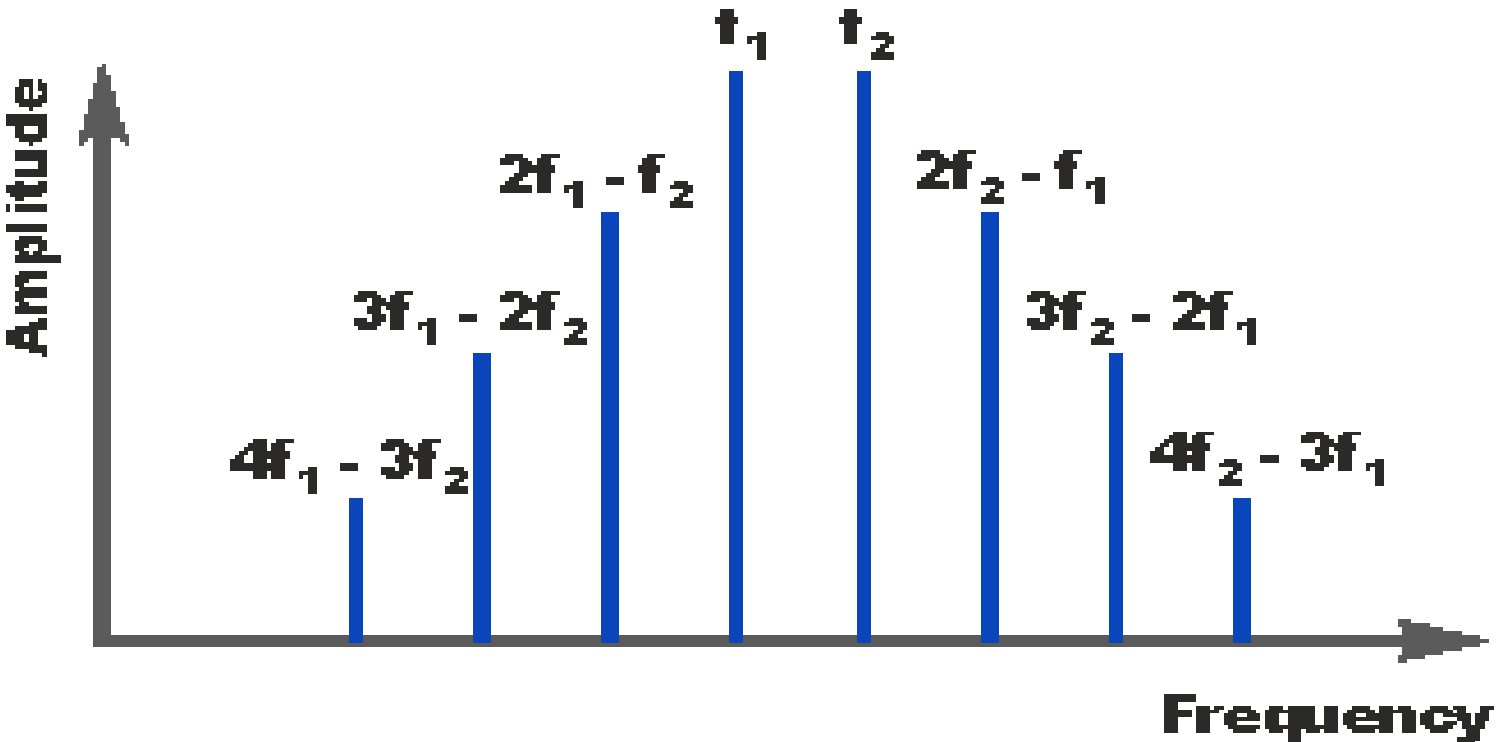
- Total Harmonic Distortion (THD) is the ratio of quadratic sum of the RMS values of all the higher harmonics to the RMS value of fundamental.

$$THD = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \ldots + V_n^2}}{V_1}$$

Correlated Noise- Intermodulation Distortion

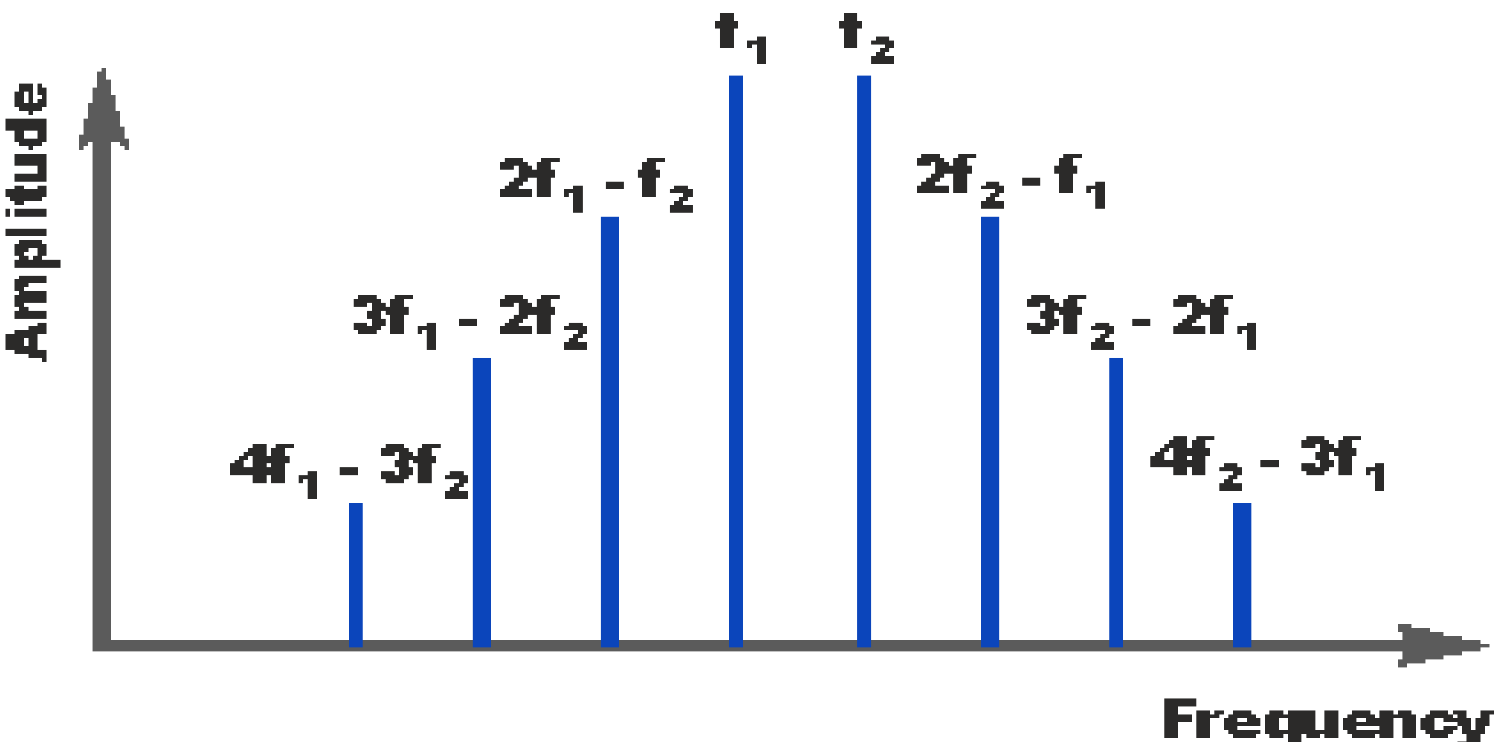
- Generation of unwanted sum and difference frequencies, when two signals are mixed in non-linear device.

- Sum and difference frequencies are called cross products and can interfere with the signal.



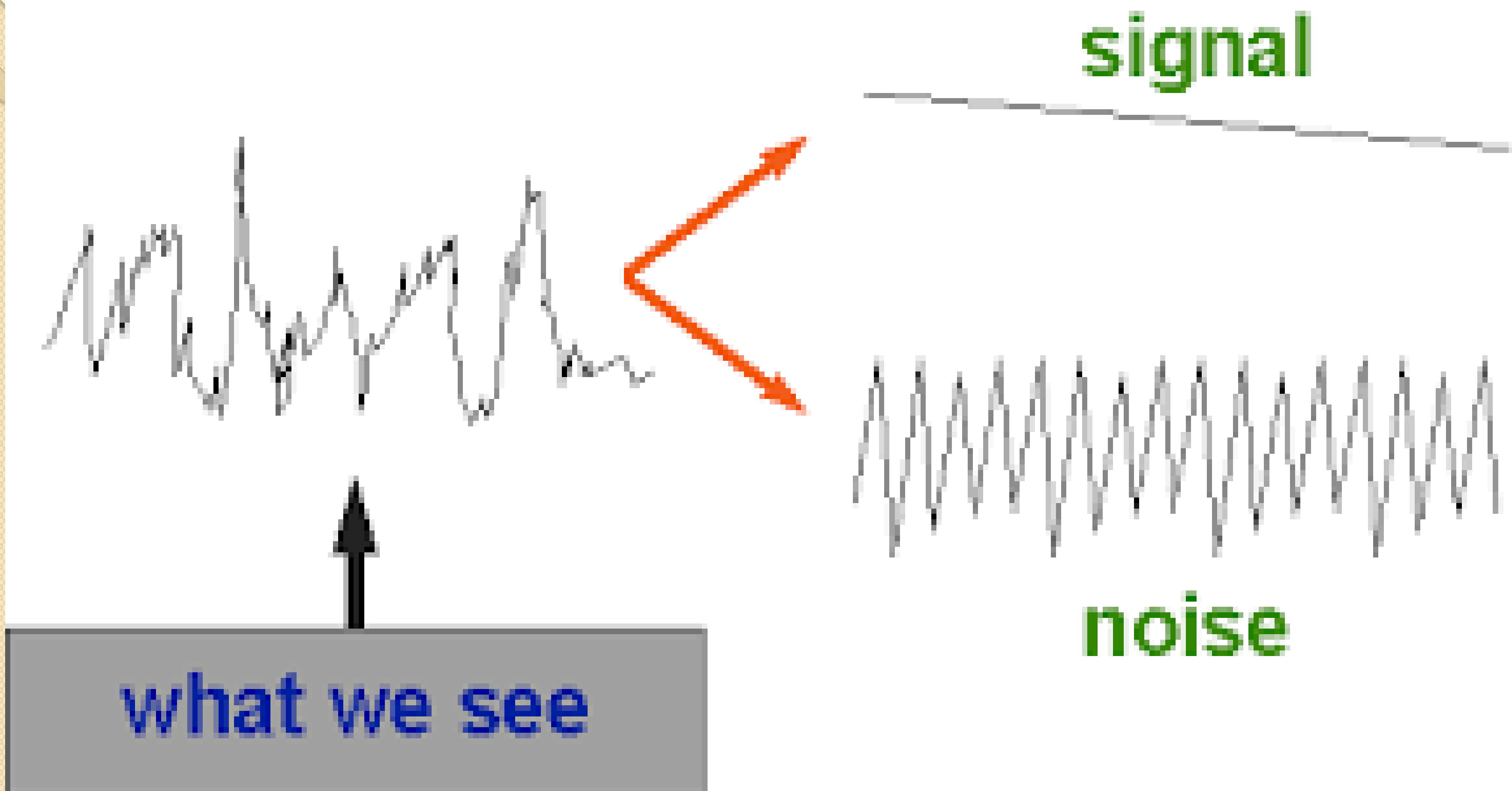
Correlated Noise- Intermodulation Distortion

- Cross products = mF1+/-nF2

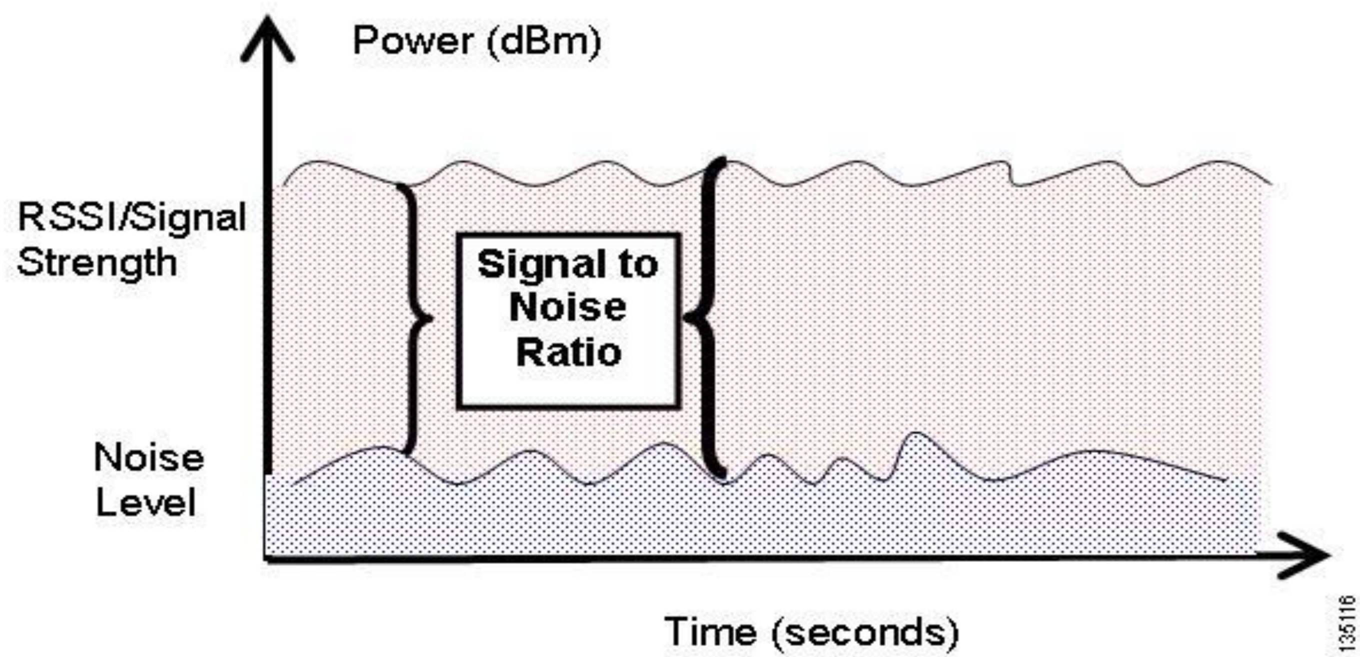


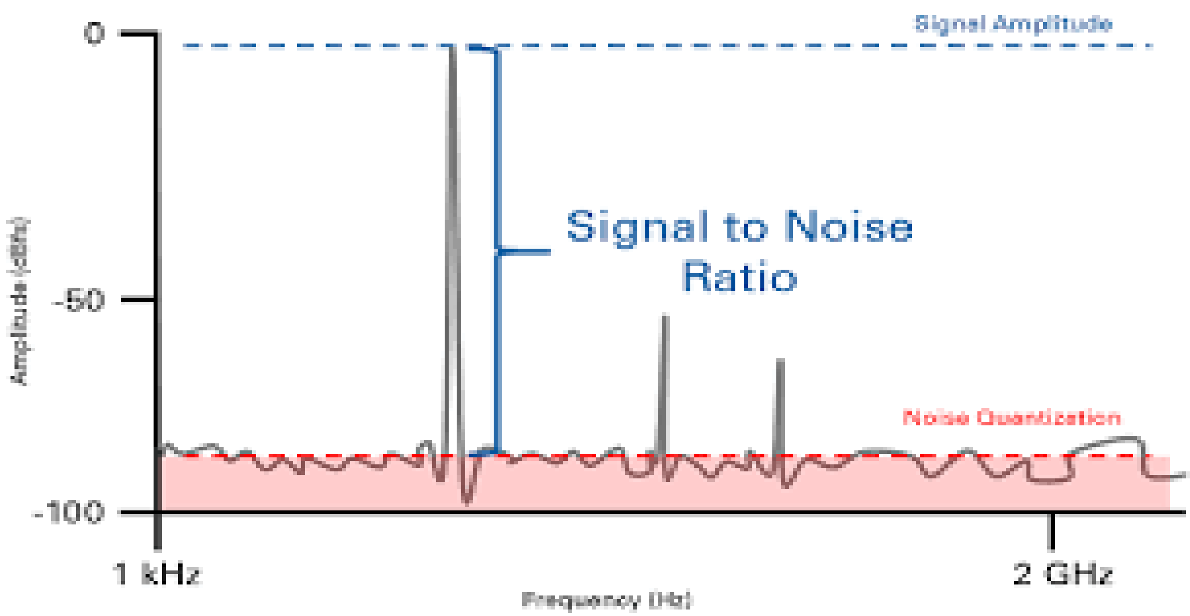
Signal to Noise power ratio

What we observe can be divided into:



Signal to Noise power ratio





Signal to Noise power ratio

$$SNR = \frac{P_{signal}}{P_{noise}}$$

$$SNR_{dB} = 10 \log_{10}\left(\frac{P_{signal}}{P_{noise}}\right)$$

$$N_{dB} = 10 \ \log_{10}\left(\frac{V2^2}{V1^2}\right)$$

$$\boxed{N_{dB} = 20 \ \log_{10}\left(\frac{V2}{V1}\right)}$$

Note:
Decibel scale to handle small and large vales of SNR. E.g. Signal of 1mW and 100 watts differ by a factor of 100,000

But in dB, it differs by 50 dB.

SNR Example

- Strength of transmitted signal is 10mW
- Noise power affecting the received signal is 1mW

- SNR (linear) = 10mW/1mW = 10

- SNR (dB) = 10logSNR (linear) = 10 dB

- **Interpretation:**

  - **SNR = 10 dB** indicates signal power is 10 times greater than the noise power (reliable communication)

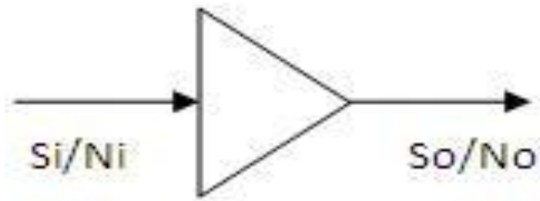
  - SNR (<0dB): Noise power is more → poor communication

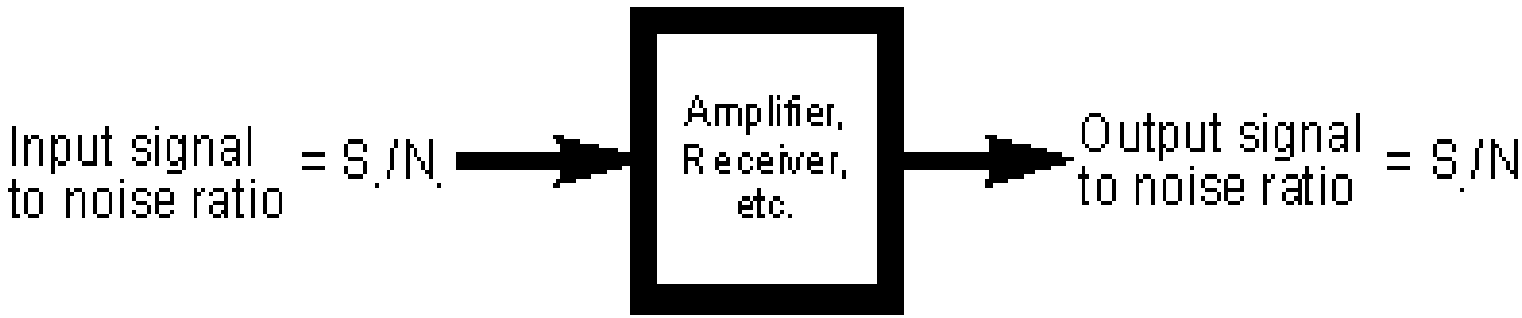
  - Moderate SNR (10 to 20 dB) → sufficient for voice or data transmission

  - High SNR (>30 dB) → near-perfect communication, low error rates, suitable for high-quality audio or video streaming

Noise Factor(F) and Noise Figure (NF)

Noise Factor = F= (Si/Ni)/ (So/No)





$$\text{Noise figure} = 10 \log_{10} \frac{S_i/N_i}{S_o/N_o}$$

"Noise Figure indicates how much S/N ratio deteriorates as a waveform propagates from input to output"

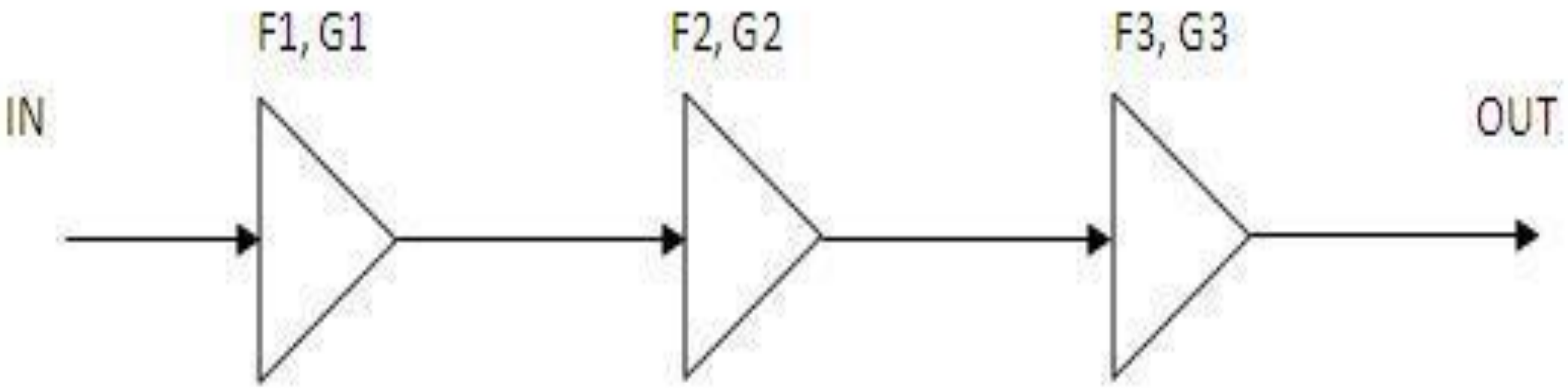
Noise Factor(F) and Noise Figure (NF)

- "Noise Figure indicates how much S/N ratio deteriorates as a waveform propagates from input to output"

- Noise Figure of 6dB indicates SNR at output is 6dB less then input.

- For noiseless circuit, noise factor is 1 and hence noise figure is 0 dB.

Noise Figure of cascaded amplifiers



Overall Noise factor,

$$F_n = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots + \frac{F_n - 1}{G_1 G_2 \ldots G_{n-1}}$$

Cascaded Noise Figure (dB) = 10 Log(Fn)

Noise Temperature

Equivalent Noise Temperature (T_e)

- Hypothetical value that cannot be directly measured.
- To indicates the reduction in the SNR a signal undergoes as it propagates through a receiver.
- The lower Te is the better quality of a receiver.

$$T_e = T(F-1) \qquad F = 1 + \frac{T_e}{T}$$

List of Formuals

1) P= KTB

2) T (Kelvin)=T(degree)+273

3) P (dBm)=10 log (KTB/1mW)

4) Vn= $\sqrt{4RKTB}$

5) THD formula

6) Te=(F-1)T

7) F$= 1 + \frac{T_e}{T}$

Example 1

- Consider a LNA in a communication system, which amplifies a weak signal. The input signal has an SNR of 20 dB and output signal has an SNR of 15 dB after passing through amplifier. Calculate noise factor and noise figure

SNR (dB)= 10 log SNR (linear)

SNR (linear) = antilog (SNR(dB)/10)

Example 1

- Noise Factor = 3.16 (Linear ratio)
- Noise Figure = 5.015 dB
- Interpretation: System introduces the additional noise that reduces the SNR by about 5 dB.
- Higher the noise figure → worst system performance
- Lower the noise figure → desired (minimal noise while amplifying signal)